

Instructions

- The solutions MUST be done in L^AT_EX and handed as pdf. No handwritten assignments will be accepted.
- Submit the assignments in time via online form.
- For python based problems, provide the codes as well as any generated figure. Seek your instructor's or peers' help if you need it. Responses during the viva that amount to: '...that's what the AI gave me', would result in a 0.
- Equations must be numbered and properly formatted.
- Expect some simple questions from the readings/viewings in the exam as well.
- Your final marks in the assignment will be contingent on the viva.

Vector Calculus

1. **Vector Fields** For the vector field

$$\mathbf{A}(x, y, z) = x^2y \hat{\mathbf{x}} + yz^2 \hat{\mathbf{y}} + zx^2 \hat{\mathbf{z}},$$

calculate $\nabla \cdot \mathbf{A}$ and $\nabla \times \mathbf{A}$. and state whether the field is solenoidal or irrotational.

2. **Vector Calculus Product Rules** Take any (non-constant) fields of your choice and verify the six product rules of vector calculus given at the end of Griffiths. However, if you can prove them in general, that would be better, and you need not choose fields.

$$\nabla(fg) = f \nabla g + g \nabla f, \quad (1)$$

$$\nabla(\mathbf{A} \cdot \mathbf{B}) = \mathbf{A} \times (\nabla \times \mathbf{B}) + \mathbf{B} \times (\nabla \times \mathbf{A}) + (\mathbf{A} \cdot \nabla) \mathbf{B} + (\mathbf{B} \cdot \nabla) \mathbf{A}, \quad (2)$$

$$\nabla \cdot (f \mathbf{A}) = f(\nabla \cdot \mathbf{A}) + \mathbf{A} \cdot (\nabla f), \quad (3)$$

$$\nabla \cdot (\mathbf{A} \times \mathbf{B}) = \mathbf{B} \cdot (\nabla \times \mathbf{A}) - \mathbf{A} \cdot (\nabla \times \mathbf{B}), \quad (4)$$

$$\nabla \times (f \mathbf{A}) = f(\nabla \times \mathbf{A}) - \mathbf{A} \times (\nabla f), \quad (5)$$

$$\nabla \times (\mathbf{A} \times \mathbf{B}) = (\mathbf{B} \cdot \nabla) \mathbf{A} - (\mathbf{A} \cdot \nabla) \mathbf{B} + \mathbf{A}(\nabla \cdot \mathbf{B}) - \mathbf{B}(\nabla \cdot \mathbf{A}). \quad (6)$$

Python – Verify the above relations by hand as well as using code.

3. **Integral Theorems**

State Gauss's divergence theorem and Stokes' theorem. Use each theorem to convert the corresponding surface or volume integral into its equivalent line or volume integral.

Electrostatics

4. **Electric Potential**

Starting from $\mathbf{E} = -\nabla V$, derive the electric field of a point charge from its potential

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \frac{q}{r}.$$

Then show that the potential due to a continuous charge distribution can be written as

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d\tau'.$$

5. **Divergence, Curl, and Electrostatics**

For an electrostatic field, explain why $\nabla \times \mathbf{E} = 0$. Show that this implies that the work done in moving a charge between two points is independent of the path taken. Explain the physical significance of the result.

Python-Based Problems

6. Coulomb's Law Calculator

Python Skills: Variables, scientific notation, defining functions, and basic math operations.

Write a Python script that calculates the electrostatic force between two point charges.

- Define a variable for Coulomb's constant, $k = 8.99 \times 10^9$.
- Write a function `coulomb_force(q1, q2, r)` that takes the magnitude of two charges and the distance between them, returning the force.
- Use the function to calculate the force between an electron and a proton separated by the Bohr radius ($r = 5.29 \times 10^{-11}$ m).
- Print the result using an f-string to format the output nicely.

7. The $1/r^2$ Drop-off

Python Skills: Importing modules, `numpy.linspace`, element-wise array operations, and basic 1D `matplotlib` plotting.

Visualize how the electric field of a $1\text{ }\mu\text{C}$ point charge decays with distance.

- Import `numpy` and `matplotlib.pyplot`.
- Create an array of distances r from 0.1 to 5.0 meters using `np.linspace`.
- Calculate the electric field array E using $E = k \cdot q/r^2$. (Note: Remember that NumPy calculates the whole array at once without a loop).
- Plot E versus r .
- The plot must include: an x-axis label, a y-axis label, a title, and a grid.

8. 1D Superposition Principle

Python Skills: for loops, lists/tuples, and accumulation variables.

Calculate the net electric field at the origin ($x = 0$) due to a line of discrete charges placed along the x-axis.

- Define a list of tuples representing charges and their x-positions: e.g., `charges = [(1e-6, 2.0), (-2e-6, 4.0), (1.5e-6, 5.5)]` where each tuple is (charge, position).
- Initialize a variable `E_net = 0`.
- Loop through the list of charges. Inside the loop, calculate the electric field contribution of each charge at $x = 0$ and add it to `E_net`. Remember to account for the sign of the field (direction) based on the charge's position relative to the origin.
- Print the final net field.

9. The 2D Potential Matrix

Python Skills: `numpy.meshgrid`, 2D arrays, avoiding division by zero.

Before visualizing 2D fields, we must calculate the underlying 2D grid of values.

- Create a 1D array of x-coordinates from -5 to 5, and a 1D array of y-coordinates from -5 to 5.
- Use `X, Y = np.meshgrid(x, y)` to create 2D coordinate matrices.
- Calculate the distance matrix $R = \sqrt{X^2 + Y^2}$.
- Calculate the potential matrix $V = k \cdot q/R$ for a point charge at the origin.
- To avoid the singularity at the origin where $R = 0$, add a small softening parameter: $R = \sqrt{X^2 + Y^2 + \epsilon^2}$ where $\epsilon = 0.1$.
- Print the shape of the resulting V matrix and its maximum value.

Readings/Viewings

- Read the attached document on the history of electromagnetism and comment in particular on the history of electromagnetism.
- Watch <https://www.youtube.com/watch?v=z1uOsg2-LTA>